Module 6 – Python Programming for Cyber Security

Syllabus:

Duration: 120 hours (40 theory hours + 50 lab hours+30 Self Learning Hours)

Objective: To introduce the student to Python programming concepts

Prerequisites: Knowledge of any programming language like C / C++

Evaluation: 100 Marks (Theory exam – 40%, Lab exam – 40%, Internals & Mini-project – 20%)

Textbook:

- Learn Python the Hard Way, Zed A.Shaw, Pearson
- Let us Python-Yashavant Kanetkar, Aditya Kanetkar

References:

- Introduction to Computer Science using Python, Charles/ Wiley
- Python Power!: The Comprehensive Guide
- Python Crash Course: A Hands-on, Project-Based Introduction to Programming
- Beginning Programming with Python For DummiesLearning Python by: Fabrizio Romano
- Python Projects by Laura Cassell , Alan Gauld / Wiley
- Python Cookbook by David B. Brain K. Jones / Shroff / O'reilly Publisher
- Head First Python by Paul Barry / Shroff / O'reilly Publisher
- Professional Iron Python by John Paul Muller / Wiley India Pvt Ltd
- Beginning Programming with Python for Dummies by John Paul Muller / Wiley India Pvt Ltd
- Python for CyberSecurity by Howard E Poston III/ Wiley India Pvt Ltd

Session 1:
Lecture

- Introduction to Python

- Python Installation
- Python Fundamentals
- Data types, Variables, Literals

Lab

- Python Installation
- Familiarizing Python IDE

Session 2 & 3:

Lecture

- Input and Output in Python
- Mutable and Immutable types
- Operators in Python
- Type Casting
- if, if-else, if – elif statements
- Nested if statement
- range() function, in and not in operators

Lab

- Programs to read and print values
- Write a program to find the area of a circle
- Write a program to find the simple interest
- Write a program to check whether a number is odd or even
- Write a program to find the largest of 3 numbers

Session 4:

Lecture

- Iteration / looping statements - for loop, while loop
- Loop else statement
- Jump statements – break and continue
- Nested loops

Lab

- Write a program to find the factorial of a number
- Write a program to reverse a number
- Write a program to check whether a number is prime or not
- Write a program to demonstrate nested loops

Session 5:

Lecture

- Strings
- String operators- concatenation, replication, membership, comparison
- String slices
- String functions and methods

Lab

- Write a program to read a string and print the number of uppercase, lowercase characters
- Write a program to input a line and print the number of vowels in it
- Write a program to reverse a string
- A pangram is a sentence that contains all the letters of the English alphabet at least once, for example: The quick brown fox jumps over the lazy dog. Your task here is to write a program to check a sentence to see if it is a pangram or not.

Session 6 & 7:

Lecture

- Lists – Creating Lists
- List operations – Joining lists, List replication
- Slicing Lists, List functions and methods
- Nested Lists
- Two Dimensional lists
- Tuples- Creating tuples
- Tuple functions and methods
- Unpacking tuples

Lab

- Write a program to find the greatest element of a list
- Write a program to delete all the occurrences of a number in the list
- Write a program to find the sum of all odd numbers in a list
- Write a program to create a tuple that stores the first 9 terms of the Fibonacci series
- Write a program to print the minimum and maximum value of a tuple

Session 8:

Lecture

- Dictionaries- Creating a dictionary
- Accessing elements of a dictionary
- Traversing a dictionary
- keys() and values() functions
- Adding, updating and deleting elements of a dictionary
- Dictionary functions and methods

Lab

- Create a dictionary of odd numbers between 1 and 10, where the key is the decimal number and the value of the corresponding number in words
- Write a program to enter names of employees and their salaries as input and store them in a dictionary
- In cryptography, a Caesar cipher is a very simple encryption techniques in which each letter in the plain text is replaced by a letter some fixed number of positions down the alphabet. For example, with a shift of 3, A would be replaced by D, B would become E, and so on. The method is named after Julius Caesar, who used it to communicate with his generals. ROT-13 ("rotate by 13 places") is a widely used example of a Caesar cipher where the shift is 13. In Python, the key for ROT-13 may be represented by means of the following dictionary:

```
key = {'a':'n', 'b':'o', 'c':'p', 'd':'q', 'e':'r', 'f':'s', 'g':'t', 'h':'u', 'i':'v', 'j':'w', 'k':'x', 'l':'y', 'm':'z', 'n':'a', 'o':'b',
  'p':'c', 'q':'d', 'r':'e', 's':'f', 't':'g', 'u':'h', 'v':'i', 'w':'j', 'x':'k', 'y':'l', 'z':'m', 'A':'N', 'B':'O', 'C':'P', 'D':'Q',
  'E':'R', 'F':'S', 'G':'T', 'H':'U', 'I':'V', 'J':'W', 'K':'X', 'L':'Y', 'M':'Z', 'N':'A', 'O':'B', 'P':'C', 'Q':'D', 'R':'E',
  'S':'F', 'T':'G', 'U':'H', 'V':'I', 'W':'J', 'X':'K', 'Y':'L', 'Z':'M'}
```

Your task in this exercise is to implement an encoder/decoder of ROT-13. Once you're done, you will be able to read the following secret message:

Pnrfne pvcure? V zhpu cersre Pnrfne fnynq!

Note that since English has 26 characters, your ROT-13 program will be able to both encode and decode texts written in English.

Session 9 &10:

Lecture

- User defined functions – Defining functions
- Arguments and Parameters
- Positional / Required arguments, Default arguments, Keyword (or named) arguments
- Returning values from functions
- Scope of variables
- Mutability / Immutability of Arguments / Parameters and Function calls

Lab

- Write a function that receives two string arguments and check whether they are same length strings
- Write a function to find the volume of a box with appropriate default values for its parameters
- Write a function that takes a number as argument and calculates cube of it. If there is no value passed to the function, the function should find the cube of 2

- Define a function overlapping () that takes two lists and returns True if they have at least one member in common, False otherwise.
- Write a function find_longest_word() that takes a list of words and returns the length of the longest one.
- Write a function filter_long_words() that takes a list of words and an integer n and returns the list of words that are longer than n

Self Learning: (SL for Sessions 1-10)

- Write a program to print Unicode characters.
- Write a program to print system path variables.
- Write a program to show variable reassignment.
- Write a program to calculate average of three numbers.
- Write a program to show type conversion errors.
- Write a program to convert string to integer.
- Write a program to calculate total and percentage.
- Write a program to check leap year.
- Write a program to simulate calculator operations.
- Write a program to find roots of quadratic equation.
- Write a program to check valid password rules.
- Write a program to find sum of digits.
- Write a program to print multiplication table.
- Write a program to check palindrome string.
- Split string into words.
- Validate email format.
- Add elements to dictionary and Traverse it.

- Write function to reverse string.

Session 11 & 12:
Lecture

- Using Python libraries and modules
- Importing modules in a program
    - Importing entire module
    - Importing selected objects from a module
- Using Python Standard library' s functions and modules
    - Using builtin functions
    - Random module
- File handling in Python
- Opening and closing files
- File Access modes
- Reading from files- read(),readline()
- Writing to a file - write(), writelines()
- flush() function

Lab

- Write a program to read lines from a text file and display all the words that has less than 4 characters
- Write a program that append the contents of one file to another
- Write a program that read the contents of a text file and count the number of lines that start with 'A'

Session 13:
Lecture

- What are Exceptions
- Default Exception Handling and Exception hierarchy
- Exception handling using try – except
- try with multiple except blocks
- finally block
- else block with try, except, finally

Lab

- Trying different programs with exception handling keywords

Session 14 & 15:
Lecture

- Classes and Objects

- self variable
- Instance variables and methods
- Class variables and Methods
- Operator Overloading
- Inheritance

Lab

- Create a class Employee which contains the details of an Employee. Create an instance of the class and call its methods
- Programs to demonstrate inheritance and operator overloading

Session 16:

Lecture

- scapy library
- ipaddress module
- dns module
- dnslib library

Lab

- Familiarising functions in the above modules

Session 17 &18

Lecture

- Beautiful Soup library
- telnetlib module
- win32evtlog module
- os module
- winreg module
- wmi module

Lab

- Familiarising functions in the above modules

Session 19 & 20:

Lecture

- shutil module
- win32 clipboard module
- libratom library
- Cryptography using Pycrypto

Lab

- Familiarising functions in the above modules

Session 21 & 22:

Lecture

- WinReg
- PyPDF
- Imaging Library
- Dpkt

Lab

- WinReg to Read the Windows Registry
- USING PYTHON TO RECOVER DELETED ITEMS IN THE RECYCLE BIN
- Using PyPDF to Parse PDF Metadata
- Reading Exif Metadata from Images with the Python Imaging Library
- Parsing Firefox Sqlite3 Databases with Python
- Using Dpkt to Parse Packets

Self Learning: (SL for Sessions 11-22)

- Demonstrate name conflicts when importing multiple modules.
- Create a package with multiple modules and import them.
- Use pip to list installed packages programmatically.
- Simulate dice rolling using random module.
- Generate random IPv4 addresses.
- Read file line-by-line using readline().
- Write data to a file using write().
- Count number of lines, words, characters.
- Encrypt file contents.
- Decrypt file contents.
- Analyze file permissions.
- Lock files using Python.

- File integrity verification using hash.
- Create and Raise custom exceptions manually.
- Implement Multilevel inheritance.
- Implement polymorphism.
- Create custom packet using scapy.
- Validate IP addresses using ipaddress module.
- Perform subnet calculations.
- Retrieve Windows event logs using win32evtlog.
- Perform clipboard monitoring.
- Process images using PIL.
- Analyze packet captures using dpkt.
- Implement PDF metadata extractor.
- Implement Password strength checker.

ASSIGNMENTS(Theory and Lab)

Theory Assignment Questions (10)

1. **Python Fundamentals**
   Explain the features of Python programming language. Discuss variables, literals, data types, operators, type casting, and the advantages of Python for automation and cybersecurity applications.
2. **Decision Making and Iteration**
   Explain Python decision-making statements (`if`, `if-else`, `elif`, nested `if`) and looping constructs (`for`, `while`). Compare `break`, `continue`, and loop `else` statements with suitable examples.
3. **Strings and Collections**
   Compare strings, lists, tuples, and dictionaries in Python. Explain mutability, indexing, slicing, built-in functions, and common methods associated with each data structure.

4. **Functions in Python**
   Explain user-defined functions, parameter passing, default arguments, keyword arguments, variable scope, return values, recursion, and the concepts of mutable and immutable arguments.
5. **Modules, Packages, and File Handling**
   Explain the role of Python modules and packages. Describe different methods of importing modules and discuss file handling operations including reading, writing, appending, and exception-safe file processing.
6. **Exception Handling**
   Explain Python exception handling using `try`, `except`, `else`, and `finally`. Discuss the importance of custom exceptions and robust error handling in software development.
7. **Object-Oriented Programming (OOP)**
   Explain the concepts of classes, objects, constructors, inheritance, polymorphism, encapsulation, operator overloading, and method overriding with suitable examples.
8. **Python Libraries for Networking and Security**
   Discuss the purpose and applications of libraries such as Scapy, `ipaddress`, `dnslib`, Beautiful Soup, `os`, `shutil`, `winreg`, and `wmi` in networking, automation, and cybersecurity.
9. **Python for Digital Forensics**
   Explain how Python can be used in digital forensics for registry analysis, PDF metadata extraction, image metadata analysis, packet capture analysis, Windows event log processing, and forensic automation.
10. **Python in Cybersecurity Automation**
    Discuss the role of Python in cybersecurity automation, including network monitoring, log analysis, threat intelligence, malware analysis, and security tool development. Explain the benefits and limitations of automation in cybersecurity.

Lab Assignment Questions (10)

1. **Python Fundamentals**
   - Install Python and an Integrated Development Environment (IDE).
   - Write programs demonstrating variables, input/output operations, arithmetic operators, conditional statements, and loops.
   - Document the outputs.
2. **String and Collection Processing**
   - Develop programs to perform string manipulation, palindrome detection, pangram verification, list processing, tuple operations, and dictionary management.
   - Compare the performance and suitability of different data structures for various tasks.
3. **Function Development**
   - Develop Python programs using user-defined functions, recursion, default arguments, keyword arguments, and list-processing functions.
   - Demonstrate parameter passing and variable scope.

4. **File Handling**
    - Develop programs to create, read, update, append, and search text files.
    - Generate statistics including the number of lines, words, and characters.
    - Handle file-related exceptions appropriately.
5. **Object-Oriented Programming**
    - Design a Python application using classes and inheritance.
    - Implement constructors, instance methods, class methods, operator overloading, and polymorphism.
    - Demonstrate object creation and method invocation.
6. **Network Programming Libraries**
    - Develop programs using the `ipaddress`, `dnslib`, and Scapy libraries to validate IP addresses, perform subnet calculations, and inspect network packets in a controlled laboratory environment.
    - Document the outputs and observations.
7. **System Automation**
    - Develop Python programs using the `os`, `shutil`, `winreg`, and `wmi` modules to automate file management, retrieve system information, and perform basic administrative tasks.
    - Record observations and discuss practical applications.
8. **Forensic Data Analysis**
    - Develop Python scripts to extract PDF metadata, image EXIF metadata, Windows event logs, and browser-related forensic artifacts from sample datasets.
    - Generate a structured forensic report.
9. **Packet and Log Analysis**
    - Develop programs to analyze packet capture (PCAP) files and system log files using appropriate Python libraries.
    - Identify network protocols, summarize communication statistics, and present the findings.
10. **Mini Project – Python Automation Tool**
    - Design and develop a Python-based automation tool that integrates file handling, object-oriented programming, exception handling, and at least one Python library covered in the course.
    - Prepare a project report including objectives, design, implementation, testing, sample outputs, limitations, and future enhancements.